PantryPal – Smart Home Inventory Tracker

Complete Project Documentation & Lab Record Report

Python • Dictionaries • Text File Handling • HTML • CSS • JavaScript

1. Project Overview & Problem Statement

PantryPal is a lightweight home inventory tracker designed to maintain household grocery and essential-item records. It allows users to add and update product quantities, search for items, identify products that need restocking, calculate total stock value, and display available and low-stock items.

2. Technical Concepts Used in the Project

Concept	How & Why It Is Implemented	Exact Code Reference
Functions	Modularizes tasks into reusable blocks.	`def add_item():`, `def view_items():`
Dictionaries	Primary storage mapping item names to quantity & price.	`inventory[name] = {"quantity": quantity, "price": price}`
Lists	Created when splitting text file lines.	`parts = line.split(",")`
Conditional Statements	Checks menu options and low stock threshold (qty ≤ 2).	`if qty <= 2: status = "Low Stock!"`
File Handling	Reads & writes to `inventory.txt` using `open()` without JSON.	`file = open("inventory.txt", "w")`
CRUD Operations	Create, Read, Update, and Delete items.	`inventory[name] = ...`, `del inventory[name]`
Exception Handling	Catches missing files and invalid inputs.	`try...except ValueError:`, `except FileNotFoundError:`